ProDA: Prompt-guided Representation Disentanglement for Action Recognition

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Abstract

Recognizing actions in long, cluttered videos is difficult when all object interactions are compressed into a single holistic representation. This issue is especially severe for multi-label settings such as Charades, where multiple actions co-occur and compete for the same visual evidence, and for interaction-heavy benchmarks such as SportsHHI, where subtle pairwise cues determine the label. We present ProDA, a prompt-guided representation disentanglement framework that conditions spatio-temporal reasoning on user-specified actions. Given a video scene graph and a multi-hot action specification, ProDA generates a specified-action prompt and a complementary prompt, uses them to parse the video graph into action-specific and complementary sub-representations, and trains the two branches with classification, correlation-suppression, and margin-based reconstruction objectives. The resulting representation is more targeted and more interpretable than holistic graph reasoning. On the Charades validation set, ProDA reaches 71.1 mAP with RGB+SSG input and a CLIP-B/16 backbone, outperforming prior compared methods including OR2G+NL at 67.5 and LaIAR+NL at 67.4. On SportsHHI, ProDA improves the baseline from 10.69 to 13.55 mAP and raises recall at 150, 100, 50, and 20 to 92.54, 87.91, 72.69, and 47.81, respectively. Ablations show that both the dynamic prompt module and the proposed normalization are important, and prompt-conditioned readout weights exhibit emergent temporal localization without additional localization supervision.

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1 Introduction

Video action recognition increasingly operates in regimes where a clip contains multiple actors, overlapping activities, and long temporal contexts. In such settings, a single entangled representation must explain several concurrent causes at once: object state changes, pairwise interactions, background context, and distractor motions. This creates a mismatch between the representation and the task. The model is asked to answer an action-specific question, but it is trained to aggregate evidence in an action-agnostic manner.

Recent work has improved video understanding through stronger spatio-temporal backbones [4, 9, 7, 14] and through structured representations such as spatio-temporal scene graphs (SSGs) [12, 22, 28]. However, existing SSG-based action recognizers still process all actions jointly. Even when the graph representation is object-centric or relation-centric, the final latent state typically mixes evidence for different actions. This entanglement hurts both discrimination and interpretability: evidence useful for one action may be diluted by another, and it becomes difficult to explain which part of the graph supports a specific prediction.

Our starting point is that action recognition is often conditional. At inference time, a user or downstream system may care about a subset of actions, or may want to separate evidence for specified actions from the rest of the scene. This suggests replacing holistic graph aggregation with *prompt-guided disentanglement*: use an action specification to steer graph parsing and explicitly split the representation into a specified-action component and a complementary component.

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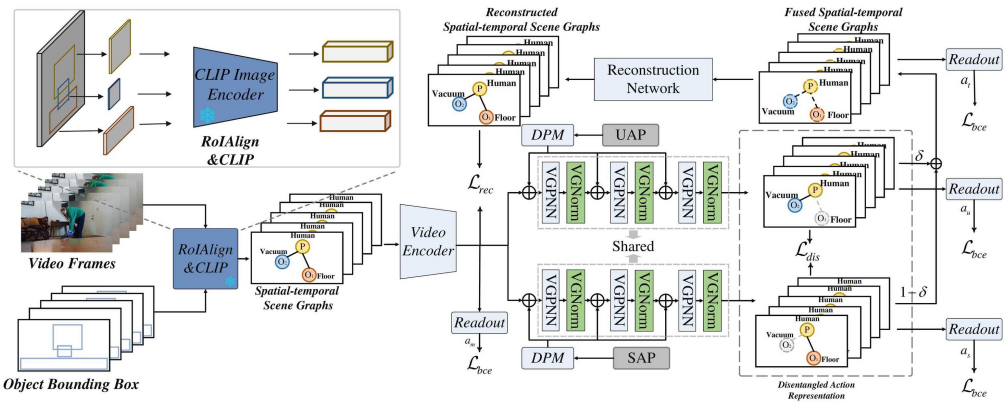

Figure 1: Overview of ProDA. A video is converted into a spatio-temporal scene graph, encoded into node and edge features, and conditioned on an action specification. The Dynamic Prompt Module produces a specified-action prompt and a complementary prompt, which guide the Video Graph Parsing Neural Network to separate the graph into action-specific and complementary representations. These branches are fused for recognition and supervised with disentanglement and reconstruction losses.

Based on this idea, we propose ProDA (**Pro**mpt-guided **D**isentanglement for **A**ction recognition), shown in Figure 1. ProDA first represents a video with an SSG and a video encoder. A Dynamic Prompt Module (DPM) maps a multi-hot action query into two prompts: a specified-action prompt (SAP) and an unspecified-action prompt (UAP), where the latter is the complementary action indicator. These prompts condition a Video Graph Parsing Neural Network (VGPNN), which parses the graph into two branches via prompt-aware message passing with temporal self-attention. A tailored Video Graph Normalization (VGNorm) stabilizes this process on variable-length video graphs. Finally, an action disentanglement objective combines classification, correlation suppression, and margin-based reconstruction so that the two branches remain distinct while preserving the information needed for recognition.

The contribution is not merely a new prompting trick. ProDA turns prompts into structural operators over a spatio-temporal graph. Prompt conditioning decides which subgraph evidence should be amplified, which should be relegated to the complementary branch, and how the final representation should be fused. This directly targets the failure mode of entangled evidence in multi-action videos.

This paper makes four contributions:

- We introduce a prompt-guided action disentanglement framework that separates an SSG into specified and complementary action sub-representations conditioned on a multi-hot action query.
- We propose VGPNN and VGNorm, which combine prompt-aware graph parsing with temporally aware normalization for video graphs.
- We develop an action disentanglement objective that jointly enforces action prediction, low inter-branch correlation, and information-preserving reconstruction.
- We show strong empirical results on Charades [26] and SportsHHI [30], together with analyses demonstrating the importance of dynamic prompts, normalization, and prompt-conditioned localization behavior.

2 Related Work

2.1 Video Action Recognition and Structured Reasoning

Classical action recognizers rely on 3D convolutional or transformer backbones to aggregate spatio-temporal evidence directly from RGB [4, 9, 7, 27, 14]. These models are effective, but their latent representations are usually holistic. They offer limited control over which part of the scene explains which action, especially in multi-label videos.

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Structured reasoning approaches address this by representing interactions explicitly. Action Genome introduced SSGs as a compositional representation of video activities [12]. OR2G reasons jointly over object-level and relation-level graphs to model relation transitions [22]. LaIAR adds language-model guidance for interpretable action reasoning [28]. Dynamic scene graph methods such as STTran [5] and relation-centric video models such as HORT [11] further show that temporal interaction structure is valuable. Yet these methods still aggregate evidence for all candidate actions together. Our focus is different: we keep the benefits of structured reasoning, but make the graph parser explicitly action-conditional and disentangled.

2.2 Prompting for Vision and Graph Models

Prompt learning has become a standard mechanism for adapting large pre-trained models. Visual Prompt Tuning [13], conditional prompt learning [32], and visual prompting for foundation models [2] show that lightweight prompts can steer recognition behavior without rewriting the backbone. Similar ideas have been extended to graph models [8, 20]. These methods typically adapt a model to a downstream task or an instance, but they do not explicitly separate action-relevant evidence from complementary scene content.

ProDA uses prompts in a more structural way. The prompt is not only an adaptation token; it determines how the spatio-temporal graph is parsed into two complementary branches. This makes prompting a mechanism for *representation factorization* rather than only for parameter-efficient adaptation.

2.3 Disentangled Representation Learning

Disentanglement has been studied extensively in images, videos, and graphs [6, 18, 33, 29, 15, 16, 19, 31]. Video-oriented work often separates static and dynamic factors [6, 33], while graph-based methods decompose latent relations or factors [19, 31, 16]. More recent task-specific systems also exploit disentanglement for interaction understanding [17].

Two gaps remain. First, prior disentanglement work is usually not conditioned on a user-specified action query. Second, graph disentanglement methods typically target latent factors without linking them to an actionable recognition objective. ProDA addresses both issues by supervising the factorization with specified and complementary action labels and by tying the decomposition to action-conditioned video graph parsing.

3 Method

3.1 Problem Setup

Let a video clip be represented by a spatio-temporal scene graph $G = (V, E)$, where nodes correspond to actors, objects, or relation entities across time, and edges encode spatial or temporal interactions. An upstream video encoder produces node and edge features $H^{(0)} \in \mathbb{R}^{N \times d}$. We consider a multi-label action vocabulary of size C. A user-specified action query is a multi-hot vector $q_s \in \{0, 1\}^C$, and its complement is $q_u = \mathbf{1} - q_s$.

The goal of ProDA is to learn two prompt-conditioned graph representations: h_s for specified actions and h_u for complementary actions, such that they are individually meaningful, jointly informative, and useful for downstream recognition after fusion.

3.2 Dynamic Prompt Module

The Dynamic Prompt Module maps each action indicator vector to a prompt that can modulate graph parsing. In the released implementation, this module is lightweight: it projects the action indicator into a prompt bank and a context vector, then predicts prompt weights conditioned on the current graph feature. Concretely, the implementation uses a linear map from the action indicator to ten prompt slots and another projection to the feature dimension, followed by a sigmoid weighting network over the prompt slots. This yields instance-conditioned prompts with minimal parameter overhead.

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Formally, for branch $b \in \{s, u\}$, DPM produces a prompt $P_b = \mathrm{DPM}(q_b, H^{(0)})$. The SAP and UAP are generated symmetrically from q_s and q_u. This symmetry matters: the specified branch should concentrate on evidence relevant to the query, while the complementary branch should explain away the remaining action evidence instead of acting as unused residual capacity.

Dynamic prompting is preferable to static prompts here because the same action query may correspond to different graph evidence in different videos. For example, the prompt for "opening" should emphasize different object trajectories depending on whether the manipulated object is a door, a drawer, or a refrigerator. Conditioning the prompt on the current graph state makes the parser sensitive to these contextual differences.

3.3 Video Graph Parsing Neural Network

Given the prompted inputs, VGPNN parses the graph into specified and complementary branches. Each layer applies prompt-aware message passing with temporal self-attention. The design is inspired by graph parsing networks [24] and transformer-style attention [27], but adapted to video graphs and conditioned on prompts.

For branch b, the ℓ-th layer computes

$$M_b^{(\ell)}(i,j) = \phi_b\Big(h_{b,i}^{(\ell)}, h_{b,j}^{(\ell)}, e_{ij}, P_b\Big)\,, \tag{1}$$

$$\tilde{h}_{b,i}^{(\ell+1)} = h_{b,i}^{(\ell)} + \sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(\ell)} M_b^{(\ell)}(i,j), \tag{2}$$

$$h_{b,i}^{(\ell+1)} = \mathrm{VGNorm}\Big(\tilde{h}_{b,i}^{(\ell+1)}\Big), \tag{3}$$

where $\alpha_{ij}^{(\ell)}$ are temporal self-attention weights over the incident edges. The parser therefore learns both *what* to pass and *where* to pass it under the action prompt.

The SportsHHI implementation instantiates this idea inside a pairwise interaction head. RoI features for the two actors and their union box are projected and concatenated with a spatial union mask, then processed by a prompt-aware transformer. The same pair feature is parsed twice: once with the specified action token and once with its binary complement. These two outputs correspond to the two disentangled branches. This implementation detail supports the broader formulation above and shows that the prompt-guided decomposition is not tied to one specific backbone.

3.4 Video Graph Normalization

Video graphs are heterogeneous: the number of nodes, valid edges, and active action factors varies across clips. Standard normalization schemes are not ideal in this regime. BatchNorm [10] mixes statistics across unrelated graphs, LayerNorm [1] ignores graph structure, and GraphNorm [3] is not tailored to temporally evolving graph parses.

VGNorm addresses this by normalizing features at the video-graph level while keeping branch-specific affine parameters. Intuitively, it stabilizes optimization across variable graph sizes and preserves the distinction between the specified and complementary branches. The ablation in Section 5 confirms that this design is substantially better than the standard alternatives on Charades.

3.5 Fusion and Prediction

After graph parsing, ProDA produces branch-level summaries h_s and h_u. The final task feature is obtained by gated fusion:

$$g = \sigma(\mathrm{MLP}([h_s; h_u])), \tag{4}$$

$$h_t = g \odot h_s + (1-g) \odot h_u, \tag{5}$$

followed by a classifier for action prediction. The released SportsHHI implementation uses exactly this pattern: a small gating module scores the two branches, and the fused feature is fed to the final classification head, while auxiliary heads supervise the specified and complementary branches separately.

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Table 1: Multi-label action recognition performance comparison on the Charades's validation set in term of mAP. SSG:ground truth SSG. Bbox: bounding box from ground truth SSG.

model	pretrain	Modality	mAP	model	pretrain	Modality	mAP
SGFB+NL [12]	K400	RGB	44.3	SGFB+NL [12]	K400	RGB+SSG	60.3
I3D+NL [4]	K400	RGB	37.5	LaIAR [28]	K400	RGB+Bbox	63.6
OR2G+NL [22]	K400	RGB	44.9	OR2G [22]	K400	RGB+SSG	63.3
LaIAR+NL [28]	K400	RGB	45.1	LaIAR+NL [28]	K400	RGB+Bbox	67.4
VicTR(B/16) [14]	CLIP	RGB	50.1	OR2G+NL [22]	K400	RGB+SSG	67.5
VicTR(L/14) [14]	CLIP	RGB	57.6	**Ours(B/16)**	**CLIP**	**RGB+SSG**	**71.1**

3.6 Action Disentanglement Loss

The training objective combines four terms:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_p \mathcal{L}_{\text{SAP}} + \lambda_u \mathcal{L}_{\text{UAP}} + \lambda_d \mathcal{L}_{\text{dis}} + \lambda_r \mathcal{L}_{\text{res}}. \tag{6}$$

Here $\mathcal{L}_{\text{cls}}$ is the main action classification loss on the fused feature, while $\mathcal{L}_{\text{SAP}}$ and $\mathcal{L}_{\text{UAP}}$ supervise the specified and complementary branches. In the provided SportsHHI code, these are implemented as separate classification losses on the prompt-conditioned branches.

To reduce overlap between the branches, we penalize correlation:

$$\mathcal{L}_{\text{dis}} = \max(0, |\text{corr}(h_s, h_u)| - \delta_d)\,, \tag{7}$$

with a margin δ_d. The released implementation computes this term from centered branch features and uses a positive margin, rather than forcing exact decorrelation. This margin is important empirically.

To preserve completeness, a reconstruction network predicts the pre-parsing representation from the fused disentangled feature:

$$\mathcal{L}_{\text{res}} = \max\Big(0, \|R(h_t) - h^{(0)}\|_2^2 - \delta_r\Big)\,. \tag{8}$$

This prevents the model from trivially discarding complementary information. Together, the two losses encourage the specified and complementary branches to be different but still jointly sufficient for recognition.

4 Experiments

4.1 Benchmarks and Protocol

Charades. We evaluate multi-label action recognition on Charades [26], a long-form indoor activity benchmark with 157 action classes. Following the provided results, we report mean average precision (mAP) on the validation split and compare against both RGB-only and structure-aware methods. The strongest ProDA result in the supplied tables uses RGB+SSG input with a CLIP-B/16 backbone [25].

SportsHHI. We also evaluate on SportsHHI [30], which focuses on human-human interaction classification in sports videos. The supplied implementation uses an RoI-based interaction model and samples short clips around keyframes; the provided ViT-B configuration samples 4 frames with frame interval 2 and trains for 20 epochs. We report mAP and recall at 150, 100, 50, and 20 detections, matching the table protocol.

Baselines. On Charades we compare with SGFB+NL [12], I3D+NL [4], OR2G [22], LaIAR [28], and VicTR [14]. On SportsHHI we compare with STTran [5], HORT [11], SlowFast [9], ACARN [23], the official SportsHHI baseline [30], and the provided capacity-matched reimplementation.

4.2 Main Results on Charades

Table 1 shows that ProDA achieves the strongest result among the compared methods on Charades. With RGB+SSG input and a CLIP-pretrained backbone, ProDA reaches 71.1 mAP, exceeding OR2G+NL by 3.6 points and LaIAR+NL by 3.7 points. The gain is notable because both of those baselines already exploit structured or language-guided reasoning. This supports the central

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Table 2: Human-Human Interaction Classification results in SportsHHI. SportsHHI⋆ is our reimplementation based on the official SportsHHI code, with minor modifications (e.g., parameter size) to match the capacity of ProDA for fair comparison.

method	mAP	R@150	R@100	R@50	R@20	Params
STTran [5]	3.31	71.29	62.91	42.67	22.14	-
HORT [11]	3.75	78.57	67.78	50.33	26.96	-
SlowFast [9]	5.00	81.66	74.00	52.74	26.82	-
ACARN [23]	5.44	82.85	75.22	56.53	31.77	-
SportHHI [30] (baseline)	10.69	89.25	82.93	68.13	43.72	111.18M
SportsHHI⋆ (reimpletmented)	10.16 (↓0.53)	90.07 (↑0.82)	84.83 (↑1.9)	68.40 (↑0.37)	40.16 (↓ 3.56)	123.11M
Ours (baseline+ProDA)	**13.55 (↑2.86)**	**92.54 (↑3.29)**	**87.91 (↑4.98)**	**72.69 (↑4.56)**	**47.81 (↑4.09)**	123.19M

Table 3: Ablation of prompt

prompt type	a_tmAP (%)	a_mmAP (%)
SimplePrompt	68.07	57.80
DPM	**71.10**	**59.04**

Table 4: Ablation of normalization

norm type	a_tmAP (%)	a_mmAP (%)
LayerNorm [1]	68.12	57.96
BatchNorm [10]	65.39	58.20
GraphNorm [3]	68.61	58.32
VGNorm	**71.10**	**59.04**

Table 5: Ablation study of loss function on Charades dataset. ❻ denotes the baseline training only the video encoder without ProDA; ❶ denotes using ProDA with only the action classification loss.

	$\mathcal{L}_{res}$ w/o margin	$\mathcal{L}_{res}$	$\mathcal{L}_{dis}$ w/o margin	$\mathcal{L}_{dis}$	a_t mAP on Charades (%)	a_m mAP on Charades (%)
❶	-	-	-	-	67.45	57.38
❷	✓	-	✓	-	68.43	8.03
❸	-	-	-	✓	67.78	58.28
❹	-	✓	-	-	64.89	57.90
❺	-	✓	-	✓	**71.10**	**59.04**
❻	-	-	-	-	-	58.01

hypothesis of the paper: disentangling action evidence with prompts is more effective than aggregating all graph evidence into a single representation.

The comparison is also informative across modalities. Relative to RGB-only CLIP-based VicTR(L/14) at 57.6, our RGB+SSG result gains 13.5 mAP. Part of this improvement comes from the structured input itself, but the comparison against prior SSG-based methods indicates that prompt-guided parsing contributes meaningfully beyond simply adding graph features.

4.3 Main Results on SportsHHI

Table 2 shows that ProDA also improves interaction-heavy recognition on SportsHHI. Starting from the official baseline at 10.69 mAP, adding ProDA increases performance to 13.55 mAP, a gain of 2.86 points. Recall also improves consistently: $+3.29$ at R@150, $+4.98$ at R@100, $+4.56$ at R@50, and $+4.09$ at R@20. The parameter count changes only from 123.11M in the reimplemented baseline to 123.19M with ProDA, indicating that the gains do not come from a materially larger model.

The SportsHHI table contains an important detail worth reporting faithfully. The provided reimplementation is slightly below the official baseline in mAP (10.16 vs. 10.69) while improving some recall metrics. ProDA is therefore evaluated against a strong but not uniformly better reimplementation. Even with that caveat, the final gains are clear and consistent across all reported recall thresholds.

This pattern suggests that ProDA improves ranking quality rather than merely trading precision for aggressive retrieval. The largest relative gain appears at R@20, where only the highest-confidence detections are retained, indicating that prompt-guided decomposition helps the model assign more reliable scores to difficult interactions. This is plausible for SportsHHI because many classes differ by subtle body configuration or role assignment. The same interpretation is consistent with the Charades gains: separating specified evidence from complementary context is most useful when several plausible actions share the same scene.

5 Analysis

5.1 Why Dynamic Prompts Matter

The first ablation in Table 3 compares the proposed Dynamic Prompt Module against a simpler static prompt. Replacing SimplePrompt with DPM improves the main score a_t from 68.07 to 71.10 and the auxiliary score a_m from 57.80 to 59.04. We retain the a_t/a_m notation from the provided tables for

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Figure 2: Frame-level weights s predicted by the readout function for different non-distractor-injected SAPs within the same video. Different prompts emphasize different temporal regions even though no extra temporal localization supervision is used.

fidelity. The consistent improvement across both metrics suggests that dynamic prompting helps both the final prediction head and the intermediate representation used during parsing.

This result is aligned with the design in Section 3.2. A static prompt can only encode category identity, whereas the dynamic prompt can adapt to the current graph state. That matters in videos where the same action label appears through different object configurations and temporal contexts.

5.2 Why VGNorm Matters

The normalization ablation in Table 4 shows that VGNorm is crucial. LayerNorm reaches 68.12, GraphNorm reaches 68.61, and BatchNorm falls to 65.39, while VGNorm achieves 71.10. The comparison is particularly informative because GraphNorm is already a graph-aware baseline [3]. The remaining gap indicates that prompt-guided parsing in video graphs benefits from normalization that respects both graph heterogeneity and temporal structure.

5.3 Loss Design and the Role of Margins

Table 5 examines the action disentanglement loss. Several observations are important.

- Adding ProDA with only the main action classification loss raises a_t to 67.45, but the full loss reaches 71.10. The structural decomposition alone helps, but disentanglement-specific supervision is necessary for the full benefit.
- Using only the decorrelation term improves to 67.78/58.28, while using only the reconstruction term hurts the main score to 64.89/57.90. Reconstruction without explicit separation is not sufficient.
- The no-margin combination yields 68.43 on a_t but collapses a_m to 8.03. This is strong evidence that the margins in the disentanglement and residual losses are not cosmetic; they prevent over-constraining the representation.
- The baseline without ProDA reports 58.01 on the auxiliary metric, so the full method improves over both the plain baseline and the partial ablations.

Overall, the ablation supports the intended role of the action disentanglement loss: the specified and complementary branches should be separated, but only up to a point. Harder pressure to decorrelate or reconstruct can destabilize the representation rather than clarify it. This also clarifies the role of the margins: the objective behaves best as a soft allocation rule, not a hard partition. The strongest setting preserves enough shared context for fusion and reconstruction while still removing redundant evidence between branches, matching the high fused score and the collapse of a_m in the no-margin variant.

5.4 Emergent Localization from Prompt-conditioned Readout

Figure 2 and Tables 6–9 provide evidence that ProDA learns temporally focused behavior without extra localization labels. For non-distractor prompts, the frame-level AP is 85.56, 57.44, and 40.18 at

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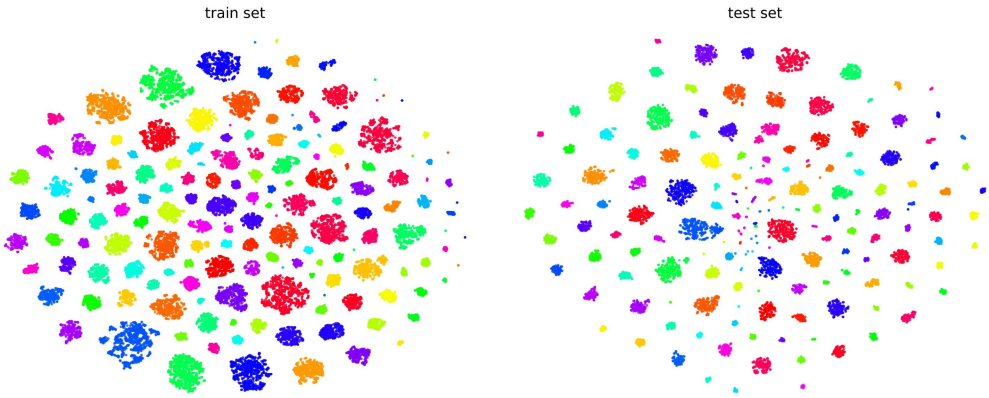

(a) (b)

Figure 3: t-SNE visualization [21] of disentangled action features across 157 Charades classes. The provided visualization suggests that prompt-guided disentanglement improves class separation in the learned feature space.

Table 6: Distractor-injected SAP

		IoU threshold 0.2	0.5	0.7	**Avg.**
θ	0.2	77.97	43.68	26.08	49.24
	0.5	64.25	33.70	20.78	39.57
	0.7	53.80	22.98	12.99	29.92
	Avg.	65.34	33.45	19.95	-

Table 7: Non-distractor-injected SAP

		IoU threshold 0.2	0.5	0.7	**Avg.**
θ	0.2	85.56	57.44	40.18	61.06
	0.5	74.44	43.43	27.75	48.53
	0.7	59.27	24.26	13.14	32.23
	Avg.	73.09	41.71	27.03	-

Table 8: Non-distractor-injected SAP. label nums indicates the length of the ground-truth label set for each video

label nums	a_s mAP	label nums	a_s mAP
1	100.00 ±0.0	9	99.80 ±0.3
2	100.00 ±0.0	10	99.45 ±0.6
3	100.00 ±0.0	11	99.62 ±0.4
4	99.66 ±0.4	12	99.10 ±1.0
5	99.80 ±0.2	13	99.56 ±0.6
6	99.82 ±0.3	14	98.98 ±1.1
7	99.98 ±0.0	15	99.19 ±1.2
8	99.88 ±0.1	16	99.80 ±0.4

Table 9: Distractor-injected SAP. label nums indicates the length of the ground-truth label set for each video

label nums	a_s mAP	label nums	a_s mAP
1	100.0 ±0.0	9	85.63 ±6.6
2	90.52 ±3.5	10	85.87 ±6.7
3	92.61 ±1.1	11	86.56 ±5.0
4	93.16 ±0.7	12	84.85 ±6.5
5	87.17 ±4.8	13	87.42 ±6.4
6	87.69 ±2.9	14	87.73 ±6.0
7	86.65 ±6.1	15	88.58 ±6.0
8	86.32 ±6.1	16	92.10 ±4.9

IoU thresholds 0.2, 0.5, and 0.7 when the readout threshold is $\theta = 0.2$, giving an average of 61.06. With distractor-injected prompts, the corresponding average is still 49.24. Averaging over θ, the non-distractor setting achieves 73.09, 41.71, and 27.03 across the three IoUs, compared with 65.34, 33.45, and 19.95 under distractors.

The robustness table by label-set size further strengthens this point. Under non-distractor prompts, the auxiliary a_s metric stays near saturation across videos with 1 to 16 labels, mostly between 99 and 100. Under distractor injection, performance remains substantially lower but still stable, generally in the mid-80s to low-90s. This indicates that prompt-guided parsing is not merely memorizing a trivial one-label regime; it remains effective even when the underlying video contains many concurrent labels.

5.5 Feature Geometry

Figure 3 visualizes the learned feature space on Charades. While t-SNE is qualitative, the provided plot is consistent with the quantitative results: classes form more separated groups after disentanglement, suggesting that action-conditioned parsing reduces overlap among classes that would otherwise

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share background or object context. This is the expected geometric consequence of separating specified evidence from complementary evidence.

Across the reported studies, a consistent pattern is that ProDA helps most when the input contains competing relational evidence rather than a single dominant cue. On Charades, the gains are strongest relative to methods that already use SSGs, which suggests that the improvement comes not only from adding structure but from partitioning that structure by action. SportsHHI shows the same effect in a different form: the parameter increase is negligible, yet both mAP and top-ranked recall improve, indicating better ordering of hard interaction hypotheses rather than simple capacity scaling. The distractor-prompt results also expose a meaningful boundary condition. Localization quality drops when irrelevant labels are injected, so the method is not invariant to prompt noise, but the degradation is gradual rather than catastrophic.

It is also useful to compare the two benchmarks through supervision density. Charades supplies dense multi-label supervision over long clips, whereas SportsHHI focuses on sparse pairwise interactions around selected actors. ProDA improves in both regimes, which suggests that the benefit is not tied to one annotation format. In dense clips, the complementary branch helps separate overlapping household activities that share objects and rooms; in sparse interaction clips, it suppresses nearby distractor motion so the queried pair receives a cleaner score.

6 Conclusion

This paper presented ProDA, a prompt-guided representation disentanglement framework for action recognition. The central idea is simple but effective: instead of asking one representation to explain every action at once, use an action-conditioned prompt to separate graph evidence into specified and complementary parts, then learn from both. Across Charades and SportsHHI, this yields better recognition accuracy than the compared baselines, with especially clear gains when structured interaction reasoning matters.

The accompanying analyses show why the method works. Dynamic prompts outperform static ones, VGNorm improves optimization over heterogeneous video graphs, the margin-based disentanglement objective is necessary for stable factorization, and prompt-conditioned readout weights exhibit useful temporal localization behavior even without explicit localization supervision. Taken together, these results suggest that prompt-guided disentanglement is a practical way to make video action representations both more discriminative and more interpretable.

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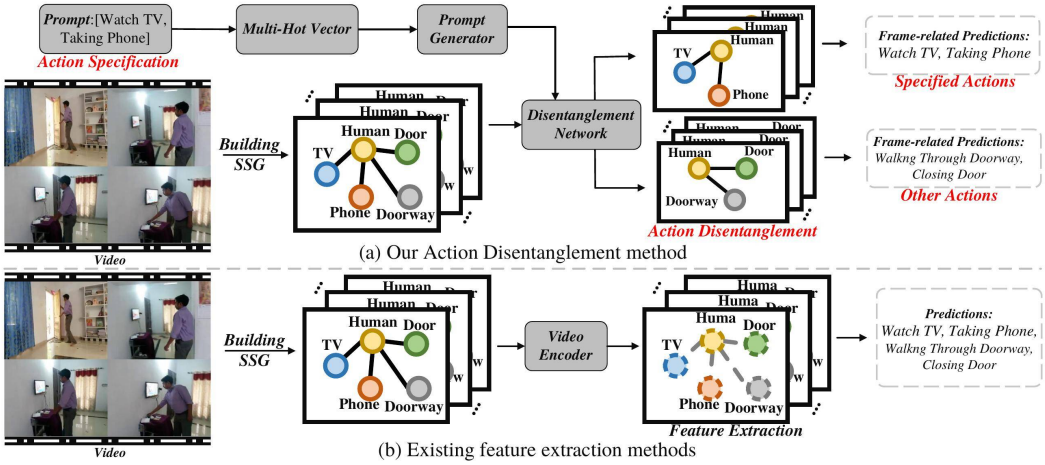

Figure 4: Conceptual comparison between existing SSG-based action recognition and ProDA. Existing methods aggregate all graph evidence jointly, whereas ProDA uses prompts to split the graph into specified and complementary branches before fusion. This makes the final decision depend on targeted evidence rather than on a single undifferentiated graph state.

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